

## **ITSC 2181 - Introduction to Computer Systems**

### **General Course Information**

#### **Course Description**

Introduction to computer system abstractions reflected in programming languages, operating systems, architectures, and networks. Topics include: overview of computer and processor architecture, instruction set architecture and introduction to assembly language, C programming, system calls, processes and process memory layout, interfaces for memory allocation and file systems, file and directory management via the command line, network architecture and protocols (such as HTTP, MAC, IP, DNS).

**This course is 4 credit hours.** If this is a core course requirement for you, you must earn at least a C to pass.

#### **Prerequisites**

ITSC 1213 with grade of C or above.

#### **Course Delivery Information**

#### **Contact Information**

Instructor contact methods and office hours vary - please see course pages for individual instructor information.

#### **Meeting Times**

All classes and lab meetings are face-to-face (in person), unless otherwise specified. *Coursework will be completed between May 18, 2026 and August 10, 2026.*

#### **Instructional Assistants**

This course has instructional assistants (IA), which are students familiar with the course material employed to assist in the instruction and conduct of the course. There are # IAs overall, who are all available to help you in your understanding of course materials. Please see Canvas for contact information for your section's IAs and the IAs of other sections. The most efficient method to ask questions or get help is to use the Canvas Discussion Board, Ninerlink, or to stop by during office hours (either in person or using Zoom). Please use email only for personal matters, e.g. medical excuses.

#### **Course Textbook**

Dive into Systems - Version 1.2, Suzanne J. Matthews, Tia Newhall and Kevin C. Webb  
You can find the [full textbook online at no cost](#).

We will use this textbook in several assigned readings. This textbook is **required**.

You only need the free online version. The textbook is also available in print and e-book form, through No Starch Press (ISBN-13: 9781718501362). *Some readings for the class are not found in the textbook. These readings will be provided in digital form, linked from the course Canvas site.*

### **Technology Requirements**

**You will need a personal computer to write, debug and test computer programs.** You will also need to be able to install and run software downloaded from the internet. Chromebooks or tablets are not sufficient.

You must have a Laptop computer meeting the [UNC Charlotte computer specifications](#).

### **Course Content Information**

#### **Course Objectives or Learning Outcomes**

At the end of this course, students will be able to do the following:

- CO1: Explain the principles, components, progression and performance challenges of computer architecture.
- CO2: Explain the instruction set architecture of a computer system and how an instruction is represented at the machine level and symbolic assembler level.
- CO3: Map major high-level language patterns into assembly/machine language notations.
- CO4: Convert numerical data between different representations including binary, hexadecimal, decimal, and other formats.
- CO5: Write, build and execute C programs.
- CO6: Use the Unix command line for file & directory management.
- CO7: Use system calls in C programs.
- CO8: Explain how hardware and software systems expose high level abstractions to lower level features in the context of system programming, networking, and architectures.

### **Course Topics**

- Principles of computer architecture and organization: Component of a computer system, Von-Neumann architecture, performance aspect of program execution on computers; abstractions for using and designing computer systems; and some of the great ideas in computer architectures such as Moore's Law.
- C programming basics and assembly format of a C program: C programming basics, variables and memory address (pointers); the relationship between arrays, pointers, and memory in C programming; numbers (binary, decimal, and hex); how a high level program is compiled, linked and executed in a computer, practice on using the command line to compile, link and execute C programs.
- Instruction set architecture using RISC-V ISA: memory and registers of the components of a computer; arithmetic and logic operation instructions; control transfer instructions; and instructions for load data from memory to register and for store data from register to memory; converting core high-level language constructs (if-else, for and while loop, variable and array references) to instruction sequence in assembly.
- Introduction to Operating Systems: purpose and responsibilities of an OS, system calls, kernel / user mode.
- Program execution and scheduling: what is a process; process states; examination / demonstration of processes in a Unix system; basic introduction to concurrent process execution and context switching.
- Memory management: the relationship between program execution and memory; program memory layout (stack, heap, etc.); memory allocation and deallocation.
- File management & command line: using the command line for basic file and directory management, including manipulation of file permissions.

## **Course Policies and Grading Information**

### **Grading Criteria**

Grading is based on multiple grading components, as detailed below:

Assignment categories, grade weights, and basic descriptions of assignments categories.

| Component                  | Weight                  | Details                                                                                                                                 |
|----------------------------|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| Exercises / Pre-Class Work | 5%                      | Activities to be completed during / after class (e.g., assignments).<br>to be completed prior to class (e.g., pre-class video quizzes). |
| Labs                       | 10%                     | Multiple lab assignments throughout the term. Usually one lab per weeks.                                                                |
| Quizzes                    | 14% each (total of 70%) | Five quizzes across the term, corresponding to each module.                                                                             |
| Final Exam                 | 15%                     | Cumulative Final Exam.                                                                                                                  |

*All percentages shown are out of the overall course grade.*

### Late Work

Late submissions outside of the original deadline in the “Exercises” group **will receive half credit and will receive no feedback but can be submitted until the last day of classes**. Assignment due dates are designed to align with the course content being taught, so staying on track is very important. Feedback on these assignments is one of the critical ways to understand your growth in the course, so the lack of feedback will heavily impact your ability to learn from assignments.

For assignments in the “Lab” group, there will be a week window after the original due date to submit the lab for full credit. After this grace period, there will be no credit given to labs submitted. We believe that the lab assignments are crucial learning opportunities for course learning objectives, and highly recommend the completion of them as early as possible.

### Academic Integrity Policy

Violations of academic integrity in this course consist of the following:

- The unauthorized submission of Generative AI code, peer's answers, or other unauthorized sources on all assignments.
- The submission of assignments from previous semesters and/or courses, regardless of whether or not the work is your own.

- Unauthorized collaboration on summative assignments, such as Labs and Exams.

Penalties for the violation of academic integrity are subject to the following penalties, as determined by the instructor's discretion. One or multiple of the following may be applied:

- Relevant Assignment(s) Receiving a Score of 0.
- Drop or Change of Final Letter Grade.
- Notification to Division of Student Affairs related to Academic Integrity.

## **Coursework**

### **Course Workload & Structure**

In this course there will be pre-class work that will fall into the exercises category based upon videos / recordings of content prior to normal course meeting time. The expectation is that students watch the videos prior to attending weekly lectures, as well as complete any associated pre-class work to validate that they understand the content from the videos. Class structure will be as follows- weekly videos and pre-class work, a lecture with a practical demonstration of the work done in the videos, an in-class assignment to evaluate student understanding, a lab session to work or get help with lab assignments, and a quiz every two weeks based on the current progression of the course. The week after each quiz will be a retake session for the quiz where you can prove you learned from the mistakes on the previous iteration. The quiz score will be a combination of these two results.

### **Attendance**

In this course attendance is not worth a grade but will be collected once at the beginning of class, and once at the end of class for one attendance credit each. This means there are 36 potential attendance credits available. If you get over 27 credits you will qualify for an additional quiz retake of your choice in the final week of the course (75% of the possible credits). Start of class attendance is taken five minutes into the course and end of class will be marked upon completion of the lecture portion.

### **Group Work**

In this course, **you may be asked to complete some in-class activities as part of a team.** Being able to work as part of a team is a very important component of your education and it is a skill that employers look for. For this reason, your cooperation is expected and appreciated.

### **Lab Work**

In this course, you will complete lab assignments, which make up a higher portion of the grade than regular programming assignments. These assignments are intended to be larger, formative assignments which can help you put course concepts into practice together, which will support your understanding for the quizzes in the course.

These lab assignments are supported by designated lab sessions that occur once a week, in which instructional assistants are available to demonstrate portions of lab assignments, explain requirements, and help you develop your code. There are a total of 5 lab assignments in this course, one for each module.

## **Quizzes**

Five in-class quizzes, all coding based with LockDown Browser. Each quiz can be retaken once one week after the original take in designated quiz sessions. The quiz score will be equal to 90% the highest attempt + 10% the lowest attempt.

**If you are caught cheating on a quiz you will receive a zero for that quiz, and not be able to retake or replace the grade.**

*Quizzes will often include content from prior modules; as such, we recommend that you engage with all of the course content frequently, and be prepared for any content covered in the course to be present on a quiz.*

Quizzes will be taken at the testing center outside of classtime. There will be a three day window to take each quiz and you are responsible for signing up for and attending the quiz session of your choice. If you do not take the quiz in this period and *do not* notify instructional team members **prior** to the quiz period starting then you will get a zero for that quiz.

There is no required attendance in this course, however if you maintain over a 75% attendance credit rate then you will qualify for an additional quiz retake of your choice in the final week of the course.

## **Final Exam Period**

The Final Exam will be cumulative and required- worth 15% of the students overall grade. It will be broken into five sections and each will serve as a potential retake for its designated quiz section, meaning section 1 serves as a retake for quiz 1. This will count towards the quiz 1 score of 10% your lowest attempt + 90% of your highest attempt with the middle score being dropped if one exists.

Please note that the contents of this document may change in order to address errors or to adjust for course progress.